

Title: Enhancing Large Vision-Language Conversational (LaViC) Recommendation Framework

Members' Names: Jason Yu, Jessie Ma, Yosef Moustafa

Abstract:

Conversational recommender systems in visually driven domains, such as e-commerce products, often struggle to capture nuanced aesthetic features through textual metadata alone. While Large Vision-Language Models (VLMs) can bridge this gap, they face significant token explosion and computational overhead when processing high-resolution images. This study evaluates the scalability of LaViC (Large Vision-Language Conversational Recommendation Framework) by investigating how varying visual data density affects the quality of knowledge distillation. Using LLaVA v1.6 (Mistral-7b) as a teacher model, we conducted a comparative analysis across the Amazon Home category, contrasting a text-only baseline against three experimental trials of increasing visual complexity: single-image distillation, dual-image distillation, and full-scale distillation utilizing the entire 18,505-image corpus. By maintaining consistent training parameters across the trials, we isolate visual density as the primary variable. Our results aim to identify the optimal balance between visual data volume and recommendation accuracy, providing insights into the efficiency of self-distillation mechanisms for deploying sophisticated multimodal reasoning in resource-constrained systems.